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A sustainable model based on genetic algorithm for garment redesign process

Manoj Kumar Paras
National Institute of Fashion Technology, Kangra, India
Lichuan Wang
Graduate School, Soochow University, Suzhou, China
Rudrajeet Pal
Faculty of Textiles, Engineering and Business, University of Borås, Borås, Sweden and
Department of Industrial Engineering and Management, University of Gävle, Gävle, Sweden, and
Daniel Ekwall
Faculty of Textiles, Engineering and Business, University of Borås, Borås, Sweden and
Supply Chain Management and Social Responsibility, Hanken School of Economics, Helsinki, Finland

Abstract

Purpose – This study proposes a garment modularization model based on an interactive genetic algorithm. The suggested model consists of extraction and identification of parts and the determination and implementation of connections. Rules and corresponding mathematical equations have been formulated for the part's extractions from the discarded products and connections for the redesigned products.

Design/methodology/approach – Sustainability entices scholars and practitioners while referring to reducing waste to control environmental degradation. One of the ways to safeguard natural resources is to increase the reuse of old or discarded products. The current study focuses on the redesign process to improve the reuse of products.

Findings – The intelligent system proposed based on the modularization techniques is expected to simplify and quantify the redesign process. The model can further help in the minimization of wastage and environmental degradation.

Originality/value – Presently, manual decisions are taken by the designers based on their memory, experience and intuition to extract and join the parts.

Keywords Sustainable design, Redesign, Fashion industry, Heuristic approach, Mathematical model

Paper type Research paper

1. Introduction

Waste generation in the garment industry has become a formidable challenge. The effect of waste on the environment has amplified in the last few years due to long-term decay and emission of toxic gases such as carbon monoxide, sulfur dioxide and methane. As a result, waste minimization in the garment industry has drawn the attention of consumers, firms and government and nongovernment agencies to make positive changes related to the issue of the environment. In other words, manufacturers and policymakers have ecological accountability to recognize the harmful impact of textile wastes and further address these issues with a resolution that interconnects the conventional design, manufacturing and end-of-life management (Cumming, 2017). The modular strategy is a vital technique to facilitate

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and interconnect traditional design and manufacturing systems with the end-of-life management. A modularized system is an intelligent approach to intervening in product design and manufacturing process to facilitate smooth disassembly for the end-of-life solutions. Various researchers have advised modularized product design methods for efficient recovery (Ma and Kremer, 2015; Wadhwa *et al.*, 2006; Wong *et al.*, 2009).

Traditional garment design and manufacturing systems were primarily operated based on human skills and capabilities, where domain knowledge of manufacturing and design was exploited for product design. However, with the advent of technology and its intervention, computer-aided design and manufacturing have influenced the garment industry in the last few years (Zhou *et al.*, 2016). The garment and fashion industry has observed most technological advancements toward improving efficiency and mass customization. An interactive genetic algorithm (IGA) is widely used to understand the product characteristics and user requirements to develop a customized and/or modular design model (Kim and Cho, 2000; Mok *et al.*, 2013). The modular design process method simplifies the disassembly and reconstruction of garments. This method can be applied to the garment industry to smoothen out the redesign process. However, field-specific knowledge regarding disassembly and reconstruction remains significant during redesigning garments (Zhou *et al.*, 2016). The garment industry illustrates diversity and complexity, but still limited research has been carried out on upstream garments (Islam *et al.*, 2021).

In the present paper, a redesign model based on an IGA is proposed to simplify garments' redesign and end-of-life management. Domain-specific knowledge of garment design and manufacturing is interpreted in mathematical formulas and model architecture to reduce the dependence on intuitive or experienced-based skills to redesign the garments. This knowledge will improve the efficiency and effectiveness of redesigning methods and contribute to the garment industry's sustainability. Thus, the aim of the current research is addressed by posing the below research questions (RQs):

RQ1. What are the techniques and methods companies follow to perform redesign activities?

RQ2. How do companies improve the existing method of garment redesign?

The current research follows the pattern of explorative study followed by mathematical and conceptual model formulation. The following sections provide a literature review of sustainability and garment redesign. Afterward, the research methodology adopted and findings from the explorative case study are elaborated. Subsequently, the mathematical model and its' application to garment redesign are presented. Finally, the research conclusion and its' implication, along with the scope for future research, are discussed.

2. Literature review

In recent decades, sustainability has drawn the attention of both researchers and industry professionals. The industry has taken multiple sustainable initiatives such as waste management, renewable energy use, carbon footprint reduction, etc. Researchers have also published numerous papers and advised different methods to improve sustainability. The concept of sustainability maintains an equitable balance between the triple bottom line, i.e. economic prosperity, environmental quality and social justice. The key to achieving a triple bottom line is the application of sustainability in business practice (Elkington and Rowlands, 1999). To inculcate sustainability in business requires sustainable products, processes and systems. All three aspects of the triple bottom line model must be addressed simultaneously to achieve sustainability (Lee and Lam, 2012). Most of the literature

published in sustainability has emphasized environmental elements. Some researchers working in sustainable operation management have identified remanufacturing as a vital method to minimize environmental impacts and manufacturing costs. Guide *et al.* (2003) have proposed various models to maximize profit and reduce the environmental burden. Pigosso *et al.* (2010) developed a remanufacturing method for the product after its end of life. The system of recycling or recovery from the waste product is also a basic method of achieving sustainability (Besiou *et al.*, 2012). The availability of waste as a raw material for remanufacturing is uncertain as recovery occurs at end of life (Linton *et al.*, 2002). Hence repair, refurbishing, part retrieval and repackaging could be alternative methods to improve sustainability. A modular structure is halfway (or kits) based on which products are designed for disassembly and reassembly. This means that parts could be detachable and repairable in the context of garments. A modular product could provide an opportunity to upgrade by changing or modifying pieces of garments. The soiled or torn parts can be replaced, and products can be used for a more extended period. Hence, a modular structure provides higher sustainability (Niinimäki and Hassi, 2011).

2.1 Redesign

Redesign requires disassembly and assembly of garments, which is more sustainable than repair. The degree of redesign varies from slight modification to complete transformation. Discarded items could be converted into valuable items with the help of redesigning process. Hence, this could contribute to economic aspects of sustainability as the redesign process can convert waste (zero value) into valuable items. The redesign provides a sustainable alternative to disposal and contributes to the environmental aspects of sustainability. The redesign process employs local designers and tailors and hence can also fulfill social aspects of sustainability. Janigo and Wu (2015) has advised redesign process as a sustainable business solution for the garment industry. Application of redesign process on discarded products could minimize cost and environmental deterioration associated with transportation and manufacturing of new garments from natural resources. Redesign arouses interest in the old products that leads to increased life of old products. Piller (2022) emphasized that redesign depends upon the disassembly effort and materials used in the product. Mono-material products and easy to disassembly products are easy to redesign. Discarded workwear, army wear, uniforms and textiles from hotels and hospitals can be transformed into new garments. This reduces dependency on natural resources by using existing products after transformation (Armstrong *et al.*, 2015).

One garment varies from others in terms of size, pattern, fabric and color of the selected products (Keith and Silies, 2015). The variation of one garment from another is the primary constraint of garment redesign. The redesign process requires excellent skill to accomplish transformation with precision by converting constraint into opportunity. However, the operation of the redesign is time-consuming and iterative. Various techniques are used to convert discarded products into sustainable development. Redesign techniques consist of three steps: ideation, reconstruction and fitting. Ideation is a creative exploration step, whereas reconstruction and fitting are implementation stages (LaBat and Sokolowski, 1999). Knowledge of pattern making and draping is essential at the implementation stage (Lapolla and Sanders, 2015). Figure 1 demonstrates the process and skill required for the redesign.

2.1.1 Ideation. Ideation is an essential step in the design process. Every individual possesses a certain amount of creativity in a particular domain. These creativities could be utilized in the fashion industry to increase sustainability and life (Lapolla and Sanders, 2015). Creativity for redesign requires passion and interest in sustainability. Inspiration for creativity and the generation of new ideas for redesign arises from the analysis of current

design and discussion of aspirational design statements. For example, the concept for redesigning a shirt can be developed using paper-made templates (patterns) for the different components, such as the sleeve, front and back. These templates are modified and re-modified multiple times to develop a unique redesign idea (Lapolla and Sanders, 2015).

2.1.2 Reconstruction and fitting. Design implementation starts with the search for raw materials from old clothes. Selected clothes are deconstructed and draped on a dress form to find the suitable application for extracted raw material. For example, polo t-shirts could be deconstructed to extract pieces of fabric. Garment manufacturing techniques such as pattern making, garment construction and draping are crucial for deconstruction and reconstruction activities (Young *et al.*, 2004). Deconstruction is one of the labor-intensive steps and must be performed carefully. The "Design for disassembly" strategy should be considered to simplify deconstruction efforts (Gam and Banning, 2011). Design for smooth disassembly, use of less diversified materials and the importance of longer stitches and permanently bound in place of fusing are suggested for smooth deconstruction (Subie *et al.*, 2009). The process of deconstruction and reconstruction made discarded garments more acceptable than direct reuse as second-hand garments. Deconstructed parts could be stitched in an assembly line or a make-through system (Young *et al.*, 2004). Abraham (2011) in her case study found that disassembly, recutting, stitching, dyeing, printing and packaging operations are performed after the reconstruction process to improve the aesthetic of garments. Embroidery or prints could be used to improve the aesthetic value of the garments. One human is different from others. So fitting is an important exercise. However, the use of elasticized waistbands eliminates fitting operation.

Further, it is hard to determine the quality and quantity of discarded garments. Therefore, it is difficult to run the continuous redesign process with a restricted and unpredictable supply of input (Paras and Curteza, 2018). Existing literature highlights the importance of redesigning activities for sustainability. Broadly, redesign process consists of the ideation and reconstruction stages. Based on the initial understanding of the redesign process, the authors have decided to start by reviewing existing literature to improve the knowledge of the garment redesign process. Subsequently, a mixed-method approach has been followed in formulating the model.

3. Methodology

A mixed approach to qualitative and quantitative techniques has been adopted to redesign activities. The study was administered in two phases in line with the research objectives and RQ posed. In the first phase, the exploratory method is followed, which involves identifying the organization engaged in redesigning used clothes, followed by in-depth interviews and on-site observations. The experimental inquiry is suitable for finding the qualitative study (Creswell, 2007) for building the theory. In the second stage, the genetic algorithm (GA) technique is utilized to optimize the redesign process. The GA technique is suitable for optimizing the design process, and this system also provides flexibility for disassembly and redesign. A GA is an artificial intelligence technique that has been used to solve problems quickly and efficiently (Paras *et al.*, 2018). Hence, the current study has first



Figure 1.
Redesign process (adapted from Paras and Curteza, (2018))

selected a case to understand the redesign process of clothes. Consequently, the GA technique has optimized the redesign process.

3.1 Research design and case selection

The case study approach has been adopted to understand the redesign process of clothes. A multiple case study is the most appropriate methodology to study complex and broad subjects (Yin, 2013). Investigation of various cases provides an opportunity to make a comparison between cases to develop a higher level of understanding. The contrast result engenders a theory-building exercise out of a comparison of cases (Eisenhardt, 1989; Eisenhardt and Graebner, 2007). The research started by searching and contacting used clothing companies located in Europe, mainly in Sweden and Romania, by email and phone. The current study has confined to the Swedish and Romanian companies that have agreed to collaborate in the recent research by sharing information. Therefore, purposive sampling has opted to select representative cases to capture the different processes involved in the redesign process, such as ideation, disassembly and reconstruction.

3.2 Description of case

The organizations chosen for the current study specialize in a different kind of redesign process. For example, organizations Redg1 and Redg2 specialize in clothes redesign, whereas organizations Redg3 and Redg4 have expertise in clothes and accessories redesign. Detail descriptions of the selected organiaztions can be found in Table I.

3.2.1 Organization Redg1. This is recycling mall working under the local municipality. The organization is working on a noble and unique concept in the form of a recycling mall. The recycling mall is located near the district energy station, where a consumer can donate used clothes and other household items. The donated clothes and other items are sorted and sold in the second-hand shop located in the same mall. The recycling mall has a collection area on the ground floor for individual residents. Redg1 does not accept industrial waste.

3.2.2 Organization Redg2. Redg2 is one of the oldest charity organizations in Stockholm. The clothes are collected through various events, sold in the shop and exported to developing countries. The charity organization Redg2 has a separate division to perform redesign activities. Recently, a redesign store was also opened at a central location near Stockholm train station. The organization does not collect through collection bins. The organization advertises collection events on their websites and request inhabitant to come and donate clothes.

3.2.3 Organization Redg3. This organization is an Iasi, Romania-based redesign organization located in the northern part of the country. The organization is under the umbrella of a 7-year-old social organization that aims to protect the environment. The organization's vision is to promote a circular economy by minimizing waste by collecting and converting discarded textiles into usable products. The positive impact on the environment is created by redefining clothes, extending the lifespan of fabric and creating new products. The customized products are made using textiles collected from local people in the bins and workshop and factory leftovers.

3.2.4 Organization Redg4. This organization is a sister company of a multinational charity organization operating in Romania. The head office of the company is located in Bucharest, and the warehouse is in Ploesti. The organization has two redesign workshops; one is located in Ploesti and the other in Iasi. The organization's primary goal is to improve the employability and life of Roma people. Various charities and social welfare projects are undertaken to achieve this goal. The organization's finance for the project is earned from the sale of second-hand clothes. Redesign clothes and accessories are exported back to Norway for sales.

Organization	Type	Description of organization
Redg1	Clothes and accessories redesign (environmental/Municipal owned)	This is a recycling mall owned by a local municipality. The organization involved collecting, sorting, refurbishing, redesigning and selling used clothes, textile, footwear, sports and construction materials
Redg2	Clothes redesign (social/not-for-profit)	This is a Swedish charity organization located in Stockholm. The organization involved collecting, sorting, redesigning and selling sorted clothes, textile, footwear and other households. Redesign activities are done through a network of satellite centers. Redesigned clothes are sold under the brand name "REMAKE"
Redg3	Clothes and accessories redesign (social/not-for-profit)	This is a Romania-based charity organization engaged in the local collection, redesign and sales of used clothes. Further, they organized different workshop types to create awareness about environment protection and upcycling
Redg4	Clothes and accessories redesign (social/not-for-profit)	This is a sister company of a multinational charity organization located in Norway. The organization received stocks of used clothes, sorting, redesign and sales through company-operated shops
Redg5	Accessories redesign (social/not-for-profit)	This is a Romania-based startup engaged in the local collection, redesign and sales of upcycled accessories. Organizations use online and offline modes to sell redesigned accessories

Note: Redg3 has recently closed shop owned by the organization

Table I.
Organizations details

3.2.5 Organization Redg5. This is also a Romania-based redesign organization registered as a social organization. The organization started its operations in 2012 in Iasi, a city in the northern part of Romania. The organization's founder's inspiration for the redesigning/upcycling came while she was on an internship in an Indian charitable organization. Based on the motivation, the organization's founder started the organization Redg5 to give a new shape to the existing waste or discarded materials. The organization aims to convert useless and discarded items into valuable products. Street banners, scrapped fabrics and old waterproof fabric (used in trucks) were converted into bags, wallets, paper folders and other accessories and stationery items. The organization is also in contract with advertising and logistics companies to receive discarded banners and truck covers.

3.3 Data collection

Semi-structured interviews and direct observations were used to collect general information about organizations and their engagement in clothes redesign. Further, the data were also collected from the organization's documents such as reports, formats and websites (Yin, 2013). Interviewees were conducted with either the founder or the managers holding top positions in the organizations, such as business managers, lead designers and project managers. The semi-structured interview intended to gain some insights from the rich experience of interviewees in the used clothing business. The interviewees were asked about the general information of the company and clothes redesigning activities undertaken by them.

The response provided by nine interviewees from five organizations were recorded and presented as anonymous. To maintain confidentiality, these are referred to as I1, I2, I3,…, I9. Table II summarizes the details of the respondents.

Case	Number of visits (direct observation)	Interviews (individual face to face)	Interviewees designation	Documents
Redg1	01	I1	Mall manager	Mall agreements documents, newspaper coverage, websites, product design, etc.
Redg2	01	I2	Manager	Patterns, modular techniques, product design, websites, social media sites, etc.
Redg3	06	I3 I4 I5	Founder Manager Designer	Patterns, work plan, product design, etc.
Redg4	02 (1 visit to the factory at Ploesti and 1 visit to the redesign workshop at Iasi)	I6 I7 I8	Country manager Factory manager Redesign manager	Inventory records, production planning charts, product design, etc.
Redg5	01	I9	Founder	Stationary and accessories pattern, product design, etc.

Table II.
Data collection details

In total, 11 interviews were conducted at the interviewee's office location. The interviews were kept short and lasted for about an hour. All the interviews were recorded and transcribed with transcribing software later on. A formal email was sent to the interviewee to give a brief about the research and the aim of the visit/interviews before conducting an on-site visit and interview. This has provided an opportunity for the interviewee to prepare data for the optimum benefits of the interview.

3.4 Data analysis

Analysis of data was undertaken in two phases. In the first phase, data analysis of semi-structured interviews aimed to find emerging themes by pinpointing, examining and observing emerging patterns within the collected data (Braun and Clarke, 2006). The emerging themes further help develop a new theory with the help of exploratory research (Eisenhardt, 1989; Eisenhardt and Graebner, 2007). The transcribed interviews and other information collected were coded in NVivo 10 qualitative analysis software developed by QSR technologies. The themes from different interviews were compared to find emerging themes and mapped with existing literature to create new theoretical knowledge as the external validity of the research contributes to the theory and practice (Yin, 2013). The articles on clothes design can be adapted to the other industry. Therefore, the findings from one study can be generalized to the different setups, too (Cho and Trent, 2006).

In the second stage, redesign process is optimized with the help of GA techniques. A modularized system is a concept that can simplify the design and manufacturing process after the end of life (Lu *et al.*, 2010). Discarded clothes after the end of life can be beaked down into parts or components to encompass new values associated with sustainability (Ma and Kremer, 2015). A modularization technique has been developed by Zhou *et al.* (2016) for flexibility and a user-friendly design process.

A GA could be applied to optimize the redesign method (Tseng *et al.*, 2008). At the selection stage (of GA), redesign parts can be selected or extracted from multiple used garments. In the next step, selected parts are reconstructed to get redesigned garments during the crossover stage. The third step is a mutation to bring minor tweaks or improvements by adding new characteristics such as print and embroidery (Kurniawan

et al., 2014). GA could be used to achieve optimal redesign from multiple garments (Kargar and Payvandy, 2015). It is difficult to achieve the best grouping for redesign through the manual hit and trial method. The heuristic approach of the GA can achieve the best redesign fit for the group (Fischer and Dagli, 2004; Junior *et al.*, 2013). In an IGA, the user (designer) can customize redesign for extracted parts, and at the same time, IGA can also suggest an optimal design to designers. Thus, IGA can receive input from the designer and interpret design information; therefore, IGA is better than the GA tool (Kim and Cho, 2000). In recent days, GAs are also used in the textile industry to solve various problems (Leo and Wallace, 2003;). Yang and Lin (2009) has used GA in the footwear industry to solve pattern problems.

A chromosome consists of a set of genes. Each gene contributes to an attribute or feature of the products. A sum of all attributes makes a unique product. Products with unique and distinctive features are parts of the population. Products suitable for extracting parts will be selected from the population (available old garments). Two to three potential products could be chosen for the redesign. A crossover operation is performed with selected products. During the crossover operation, extracted parts from different garments will be joined. When the crossover operation is completed, the mutation operation will be used to bring diversity to the redesigned garments. Some value addition operations could be performed during this stage, such as embossing print or adding embroidery to the redesigned garments.

Figure 2 demonstrates the process of GA in the garment redesign process.

The reconstructed garments may need to insert some new genes, such as trims, decorative stitches or any other value addition to make garments attractive.

A genetic model has been developed in the current study consisting of four stages. At stage I (first), the rule for the extraction of parts from used clothes is defined. The extracted parts are determined based on attributes and the value of each attribute. The parts are categorized into important and less important parts on the basis of their importance for future use. During stage II, the extracted parts will be identified for reconstruction. A binary decision variable has been set for suitable and non-suitable parts. The connection between the two parts is determined at stage III. At stage IV, the connections are implemented using a GA.

4. Case study findings

Organizations Redg1 and Redg2 focus on the improvement of the used product value. To improve the value of collected goods, Organizations Redg1 and Redg2 are involved in redesign activities. Redg1 is basically a facilitator (mall management) and tries to improve the value of old products by remaking or refurbishing, while Redg2 is redesigning new garments from old garments by following the concept of modularization. Redg2 has engaged in upcycling and established its own redesign brand and opened an exclusive store for upcycled garments. The charity organization Redg2 has a separate division to perform redesign activities. Even the organization has developed a new brand for redesigned clothes. Exclusive stores for redesigning clothes are opened at a central location in Stockholm near the Stockholm train station. In the organization Redg1, shop managers or owners are not allowed to pick items from unsorted or other sections (separate items sorted for each of the shops/stores of the recycling mall). Sometimes items are also sorted based on the requirements of the vocational design school and handed over to the school to perform redesign activities.

Organization Redg3 is a Romania-based redesign organization located in the northern part of the country. The organization Redg3 is found to be one of the most successful and economically sustainable organizations. There are five full-time employees involved in the redesign process. The organization works like any other professional organization. The organization participates in different kinds of street festivals to showcase and sell upcycled

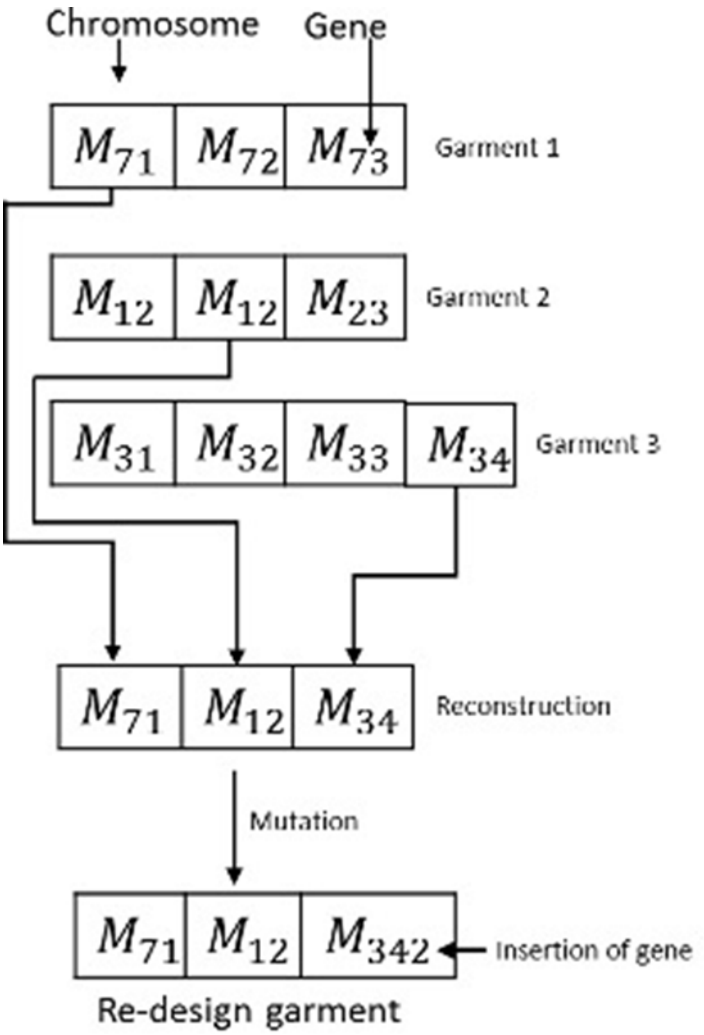


Figure 2.
GA in the garment redesign process

or redesigned products. In order to create awareness about the redesign and reuse of textile-based products, the organization Redg3 also organizes (re)design workshops at regular intervals. The designers, managers and founders of the organization Redg3 keep themselves updated about the latest redesign trends and policies. Recently, organization Redg3 has collaborated with the shop of Redg4 to receive unsold jeans for the redesigning purpose.

Organization Redg4 is a sister company of a multinational charity organization. The head office of the company is located in Bucharest, and the warehouse is in Ploesti. The organization has two redesign workshops; one is located in Ploesti and the other in Iasi. The main goal of the organization is to improve the employability and life of Roma people. Various charities and social welfare projects are undertaken to achieve the goal. The finance

for the project is earned by the organization from the sale of second-hand garments. Redesign garments and accessories are exported back to Norway for sales.

Redg5 has focused on banner and truck cover–based accessories and stationeries. The organization has collaborated with multiple advertising agencies to receive old banners. Most of the advertising companies are located in the capital city of Romania, i.e. Bucharest. Hence, the Redg5 needs to pay transportation costs to get banners and old waterproof covers (of trucks) in Iasi. Redg5 has two employees to cut, sew and pack the accessories and orders. There is no shop owned by the company, and therefore, the company has collaborated with a few shops located in different cities of Romania. The shop owners receive a fixed amount of commission for the display and sale of the products. The organization also uses an e-commerce platform for the sale of upcycled products. Besides the retail sales, the organization also receives bulk orders from the company for the events such as conferences, training, etc. Similar to Redg3, Redg5 also participate in various street markets to show its presence and sell its products.

A few designers and second-hand garment companies are making efforts to turn textile waste into cutting-edge designs. For example, in the current study, Redg1, Redg2, Redg3, Redg4 and Redg5 are engaged in the redesign process. However, the total amount and level of redesigned clothes are limited. Redesigning can be performed to repurpose garments into cushions, pillows or reconstructed apparel. For example, the sweater can be turned into a pair of pants for the child. Fabrics of jeans and trousers have a long life, and hence, they can be recut and restitched into other products. In the current study, the organization Redg3 is performing a redesign to convert used garments into different wearable garments or accessories. Most of the redesigned products are made by Redg3 on the basis of bulk orders or demand from their own shops. Redg2 has a separate redesign section and satellite center to perform redesigning activities. Useful parts from the used clothes are extracted and stored for future redesign. Redg4 and Redg1 are engaged in the redesign process for a few particular products. The redesign or upcycling could be product, process or demand based. The redesign process increases the value of the existing garments and avoids the production of new garments. Figure 3 depicts a different kind of upcycling process.

Altering, remaking and refashioning garments are other ways through which the life of garments can be extended. Specifically, the quality and appearance of the used garments in the second-hand market are improved significantly. The charities from Nordic countries send materials to Eastern Europe for recycling/upcycling. It is not financially viable to set up used clothes reprocessing units in developed countries (Carlsson *et al.*, 2015). This is also apparent from the current case study findings (especially from Romanian organizations).

5. Model development for garment redesign

A garment redesign model based on an IGA has been developed. The model includes the extraction of different parts from old/discarded garments on the basis of their future use suitability. Each of the extracted parts has different characteristics and usefulness. Some parts are highly useful while others are of less importance. Further, on the basis of demand, each part can be assembled and reconstructed in line with a GA to achieve functional and aesthetic value. Figure 4 illustrates the flowchart of the garment redesign process.

According to the redesign goal, the redesigned system searches for old garments to extract parts. This is in accordance with the selection of parents from the diverse population in the GA. The extraction of parts from old garments is similar to the selection of parent genes for the child. The garment parts are connected based on the connection constraint on the basis of domain knowledge of garment design and manufacturing. The selected parts

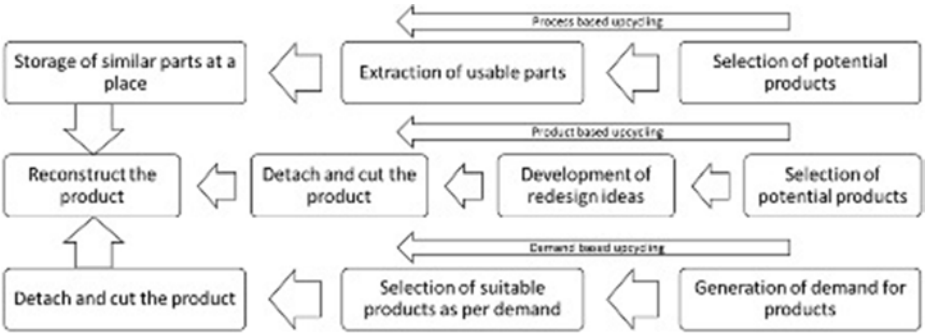


Figure 3.
Approach to the upcycling process

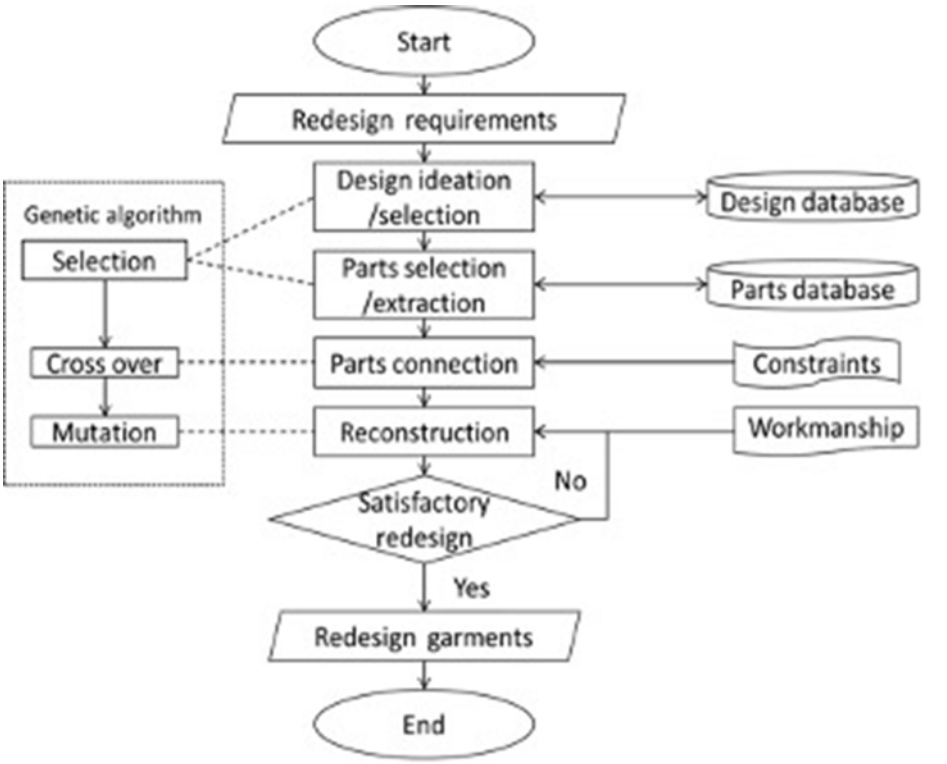


Figure 4.
GA-based redesign process for garments

are connected in line with the genes crossover of GA. The connected parts are reconstructed to achieve a unique design as mutation of crossover genes. The process of mutating and joining extracted parts should be continued until the redesign goal is achieved. This system provides flexibility to designers and customers over the current intuitive redesign process.

5.1 STAGE I: extraction of parts

Multiple parts can be extracted from an old garment, which can be classified into important and nonimportant parts. The classifications of parts are on the basis of the length and width of extracted fabrics. Important parts are those parts that are more significant for the redesign purpose. The extraction of parts should be of average size. Too small parts lead to a complicated redesign process. Information on extracted parts will be stored in the system in the form of binary variables. Below variables can be used to define the parts and their attributes mathematically:

$$M_{ij} = \{Mu, T, V\},\ i = 1, 2, 3, \ldots, n;\ j = 1, 2, 3, \ldots, n, \tag{1}$$

where M_{ij} is j^{th} portion of ith parts; $Mu = \{Mu_1, Mu_2, Mu_3, \ldots, Mu_j\}$ is the set parts extracted from M products; $T = \{t_1, t_2, t_3, \ldots, t_j\}$ is the set of attributes of parts, i.e. important parts, less important parts, etc.; $V = \{v_1, v_2, v_3, \ldots, v_j\}$ is the value set of the above attributes.

The process of extraction of parts from all available second-hand clothes is clearly explained in the above equation. The extracted parts can be stored according to their attributes T and value V.

5.2 STAGE II: identification of parts

The system has a database of garment parts after extracting parts from used garments and determining the attributes and values associated with each attribute. Based on the product design requirements, the idea for their design is developed, and the required parts are configured accordingly. The redesigned garments or accessories products required different important and less important parts. A binary decision variable has been defined to identify the requirements of parts as follows:

$$\alpha_{ij} = \begin{cases} 0 \\ 1 \end{cases} \quad i = m, l(\text{important, less important})\, j = 1, 2, 3, \ldots, n. \tag{2}$$

For garments, $\alpha_{ij} = 1$ when ij^{th} parts are identified for the configuration. when ij^{th} part cannot be configured. During the redesign process of garments, all important parts are present. Each part has a binary decision value to identify the configuration and selection.

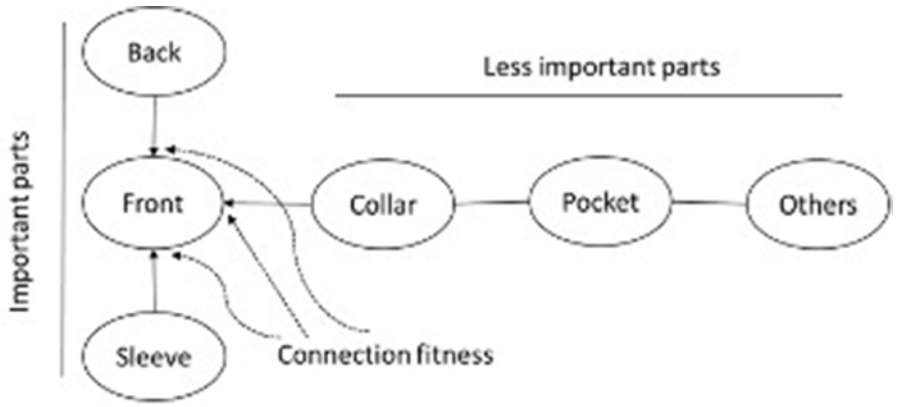


Figure 5.
Identification of parts

Generally, the garments redesigned required important parts, while accessories redesign needed less important parts. So $\alpha_{li} = 0$ for most of the cases of garments redesigned, while $\alpha_{mi} = 0$ for accessories redesigned.

Figure 5 depicts the requirements of parts for the configuration of redesigned garments and accessories. The identification process is based on importance. Important parts are necessary for garments redesigned, while less important parts are barely required. The configuration of parts is requisitely based on redesign requirements. The below picture demonstrates the suitability of important and less important parts for connection:

5.3 STAGE III: determinations of connections

The next step is parts connection after parts selection. With the simple linking of one part with others, the desired aesthetic value cannot be achieved. Patterns are required to stitch each part of the seam or junctions. The connection relationship among the interconnected parts needs to be determined for each seam. Each connecting part has a unique matching part to stitch. The matching and stitching of two parts need domain knowledge of garment design and manufacturing. Garments or accessories junctions can be defined as follow:

$$R_{ij} = \{id, M, T, V, Mu\},\ i = 1, 2, 3, \ldots, n;\ j = 1, 2, 3, \ldots, n, \tag{3}$$

where R_{ij} is j^{th} junctions of ith parts; id is the identification of connection sites or seam; M is the main part of the junctions; $T = \{t_1, t_2, t_3, \ldots, t_j\}$ is the set of attributes of junctions, i.e. shape, location, size, etc.; $V = \{v_1, v_2, v_3, \ldots, v_j\}$ is the value set of the above attributes; $Mu = \{Mu_1, Mu_2, Mu_3, \ldots, Mu_j\}$ is the set of parts which will connect to these parts.

The two parts from the original garments, M_1 and M_2, can be connected at a seam id_1 with a set of parts Mu_{12} and Mu_{21}. Each junctions or side of parts of M_1 could be connected to different parts with the help of different identified seams.

The strength or intensity of connection depends upon different connection attributes such as seam type (id), machines used to stitch seam, parts attributes (T) such as fabric composition, fabric color and junction/connection attributes value (V).

5.4 STAGE IV: implementation of connection

The implementation of connection is required after a determination of proper connection between two parts. During the implementation phase, a slight modification in the parts may be required to achieve implementation. The shape and size of two parts at each seam should be similar to become a pair. This can be achieved if attributes and value of attributes at the connection seam are equal. This can be explained with the help of the below equation:

$$\begin{cases} t_i = t_j \\ V_{Ati} = V_{Btj} \end{cases}, i = 1, 2, 3 \ldots I; j = 1, 2, 3 \ldots J \tag{4}$$

where t_i and t_j are elements of the set Mu_A and Mu_B. I and J are the total numbers of elements of a set. The set of attributes of two parts at the junction is similar, and this can be depicted by $t_{i=}t_j$. The value of variables at a junction is equal to l, which can be indicated by $V_{Ati=}V_{Btj}$. Two parts at a junction are equal in terms of size, shape and value. Most suitable parts are identified for crossover. These parts may be from different garments. The selection or identification of parts is an iterative process. Once parts are identified for the new garment G_i, the identified parts can be the crossover for reconstruction.

6. Application of model for garment redesign

Based on Section 5, a computerized model can be developed. A clothes redesigned model based on the genetic technique has been developed. The genetic method includes the extraction of different parts of old/discarded garments on the basis of their future use suitability. Each extracted part has different characteristics and usefulness. Some parts are highly useful, while others are of less important use. Then on the basis of demand, each part is assembled and reconstructed to achieve functional and aesthetic value. Figure 6 illustrates the model for clothes redesign.

On the basis of demand or redesign objectives, system operation is performed to search for suitable parts that were extracted from old or used garments. The garment parts should be selected based on attributes and value of attributes in order to achieve a higher functional and aesthetic value of redesigned garments. Extraction, selection and connections of parts required domain knowledge of garment design and construction. If garments redesign outcomes satisfy the demand requirements, then the redesign process is considered to be completed. Otherwise, more modification and reconstruction are needed to achieve the desired look. A designer can easily identify the suitable design and information about required parts with the help of a garments redesign model. This model also facilitates the designer in the parts extractions from old and used garments for optimal use. The database of design ideas and parts will enable the designers to make quick and fast decisions on the basis of previous redesigned outcomes. The garments redesign model can be utilized extensively and effectively if redesign and parts databases are updated

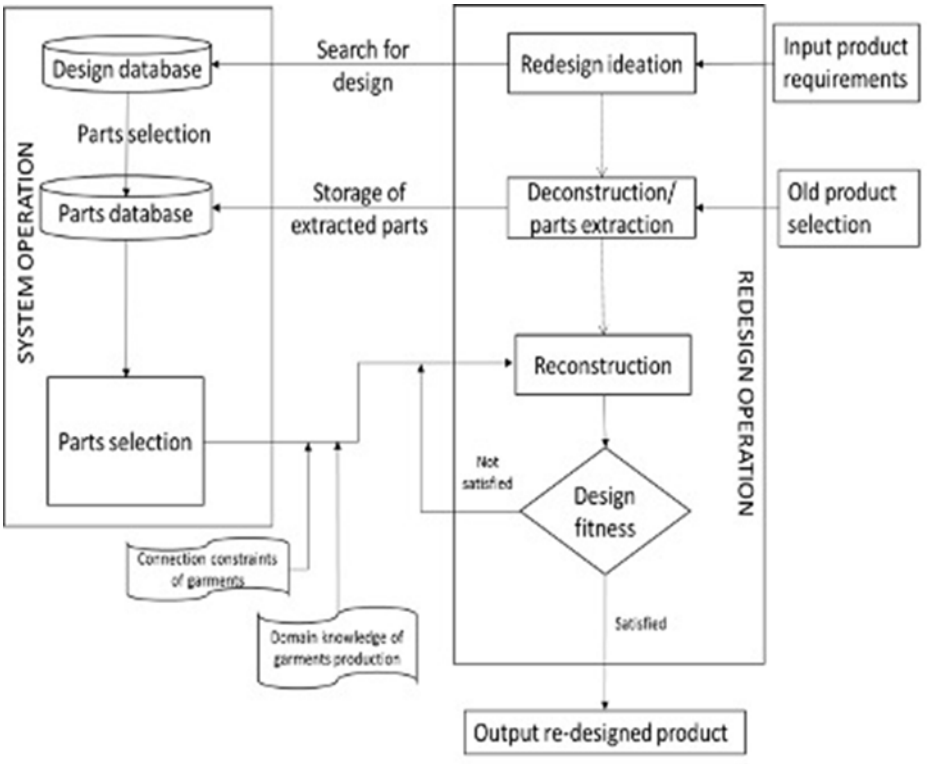


Figure 6.
Model for garment redesign

regularly. The garments redesign model based on the genetic technique could become an important tool for redesigning process. Circularity concepts such as redesign significantly contribute toward sustainability (Islam *et al.*, 2021).

A designer can easily identify the suitable design and information about required parts with the help of a clothes redesign model. This model also facilitates the designer in the parts extractions from old and used clothes for optimal use. The database of design ideas and parts will enable the designers to make quick and fast decisions on the basis of previous redesigned outcomes. The clothes redesign model can be utilized extensively and effectively if redesign and parts databases are updated regularly. The clothes redesign model based on the genetic technique could become an important tool for a redesign. The tools and systems for the redesign change consumer behavior and contribute to the longevity of the products. Furthermore, mechanical recycling leads to degenerated and downcycled products, whereas chemical recycling is used at a limited scale in the textile industry (Piller, 2022). Thus, the proposed redesign model could be considered useful.

7. Conclusions and implications

In the proposed model for garment redesign, the parts are extracted from used or discarded garments based on the redesign requirements. The extracted parts are divided into the important and less important groups. Some parts are considered important, while others less important according to their usability or suitability for the redesign. Each part has a different set of attributes and characteristics that differentiate one part from the other. The proposed mathematical definition can be used to quantify the attributes and characteristics of extracted parts. Information for each part is saved in the part database on the basis of the mathematical definition of parts. The stored parts can be identified for a redesign on the basis of design requirements. A binary decision variable can be used for the identification of the parts from the database. The value of the binary variable for the important part can be set as 1 for the apparel redesign and 0 for the accessories redesign. The mathematical symbol has been used to define parts junction and parts associated with the junction. Mathematical notation is used to clearly describe part attributes, the value of its attributes and the requirements of the junction. The system could finish redesign on the basis of domain knowledge of garment design and production. The attribute and value of the attribute of the two parts should be equal for the implementation of the connection. The formula for defined connection intensity is defined on the basis of connection attributes. A numerical value is assigned to each of the connection attributes, such as the basis of the strength of the connection.

This research theoretically advances the understanding of redesign activities by providing deep insight. Particularly, the study provides a futuristic model to optimize the redesign process based on genetic techniques. This model has several implications for the practitioners involved in the redesign activities of clothes. The existing practice of hit and trial methods to find the best match of parts for the redesign could be replaced by a computer algorithm. An algorithm can find suitable garments from thousands of diverse designs garments easily. Further computerized redesign systems will also advise practitioners on various alternatives to design aesthetically optimal clothes. The existing practice of manual clothes redesign is a complicated and time-consuming process. However, the algorithm of the computerized model of the paper will cut short the time requirements. The paper shows the pertinence of the application theory on practice. The results may contribute to influencing public policy and may bring economic and commercial impact in the long term.

This study has a number of limitations; hence, results and findings of current research should be tested more rigorously in future. For example, a lot of approximations are taken into consideration while defining binary variables. An in-depth study may be undertaken to

reduce approximation and increase the exactness of binary variables. Second, the model developed is purely conceptual in nature. However, researchers have made an effort to cover most of the factors influencing clothes redesign activities. But the model needs to be validated to test. Another limitation of the study is its focus on specific regions (i.e. Sweden and Romania) to understand the redesign activities. A more generalized understanding of redesign activities could be developed by exploring another socioeconomic region.

The limitations mentioned above provide avenues for future research. The future scope of the current research could be the integration of the CAD/CAM system with the proposed model. A two-dimensional or three-dimensional CAD technique could help to visualize the appearance of the garments. Beautiful and high-end redesigned garments can be achieved by exploring the parts database and probable connections. However, the number of parts and probable connections can be enormous. So, future research could focus on establishing the relationship between parts and design. The developed redesign model could also be applied to other industries to improve end-of-life management. The current research can be extended to develop an efficient expert knowledge system to make use of garment and intelligent system knowledge. The current research is based on pre-pandemic data, and the pandemic period should contribute to changing the practices of these organizations. Hence, future research can be conducted to test the result in the post-pandemic environment.

ORCID iDs

Manoj Kumar Paras http://orcid.org/0000-0001-7193-5362
Lichuan Wang http://orcid.org/0000-0001-8900-4128
Rudrajeet Pal http://orcid.org/0000-0003-2015-6275
Daniel Ekwall http://orcid.org/0000-0003-0871-1838

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Corresponding author
Manoj Kumar Paras can be contacted at: manoj.kr.paras@gmail.com

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